Insulin Resistance Risk Predictor using Machine Learning

INTRODUCTION:

In India, insulin resistance is a very common diagnosis. It keeps on increasing due to poor diet, lack of physical activity and irregular sleep patterns. Early detection of such risks can help in prevention.

This project aims to build a machine learning-based system that predicts the risk level of insulin resistance using lifestyle and physical parameters

PROBLEM STATEMENT :
To develop a predictive model that classifies individuals into low, moderate or high risk categories based on inputs from the user

METHODOLOGY:

Data Collection: A synthetic dataset was manually created on realistic lifestyle conditions using reliable sources

Features Used

Write as bullet points:

- **Age**
- **Waist circumference**
- **Height**
- **Physical activity**

- **Sleep duration**
- **Sugar intake**
- **Family history**
- **Stress level**
- **Water intake**

4. Data Preprocessing

Categorical values such as Low, Medium, and High were converted into numerical values using mapping techniques.

5. Model Used

A Decision Tree Classifier was used for this project. It is simple, interpretable, and suitable for small datasets. The model learns patterns from the input data and predicts the risk category accordingly.

5. Implementation

This implementation is made by using Python. Libraries used are pandas and scikit-learn. The model is trained using pickle, and later, the command line where users can put their values and get the output.

RESULTS:

It predicts the amount of risk the user holds for insulin resistance.

CONCLUSIONS:

This project demonstrates how machine learning can be applied to solve real-world health-related problems. Although simple, it highlights the importance of lifestyle factors in predicting the risk of insulin resistance.